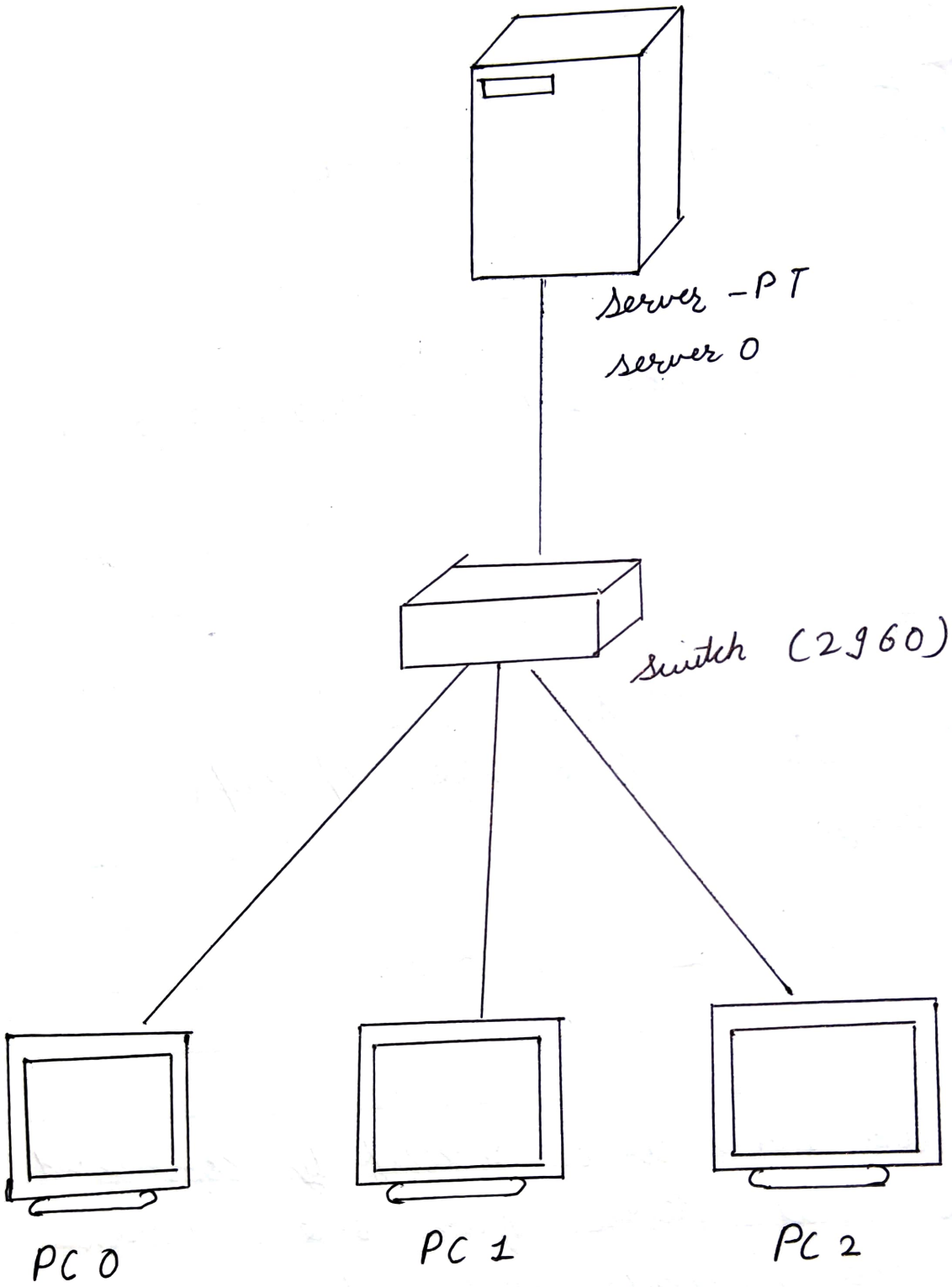

Fig :- Dedicated DHCP server

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Aim :- To configure DHCP in a network using
i) Dedicated DHCP Server
ii) Router as DHCP Server

PART A : DHCP using dedicated Server

Requirement : 1 Server, 1 Switch (2960), 3 PCs, straight through cables.

Procedure :-

1) Connect Devices
Connect all PCs and Server to Switch using straight cable

2) Configure Server
Go to : Server → Desktop → IP Configuration
- IP Address → 192.168.1.2
- Subnet Mask → 255.255.255.0
- Default Gateway → 192.168.1.1
- DNS Server → 8.8.8.8

3) Turn ON DHCP Services
Go to : Server → Services tab → click DHCP
Turn Service ON

4) Create DHCP Pool
Fill details
- Pool Name : ServerPool.
- Default Gateway : 192.168.1.1
- DNS Server : 8.8.8.8
- Start IP Address : 192.168.1.10
- Subnet Mask : 255.255.255.0

Teacher's Signature

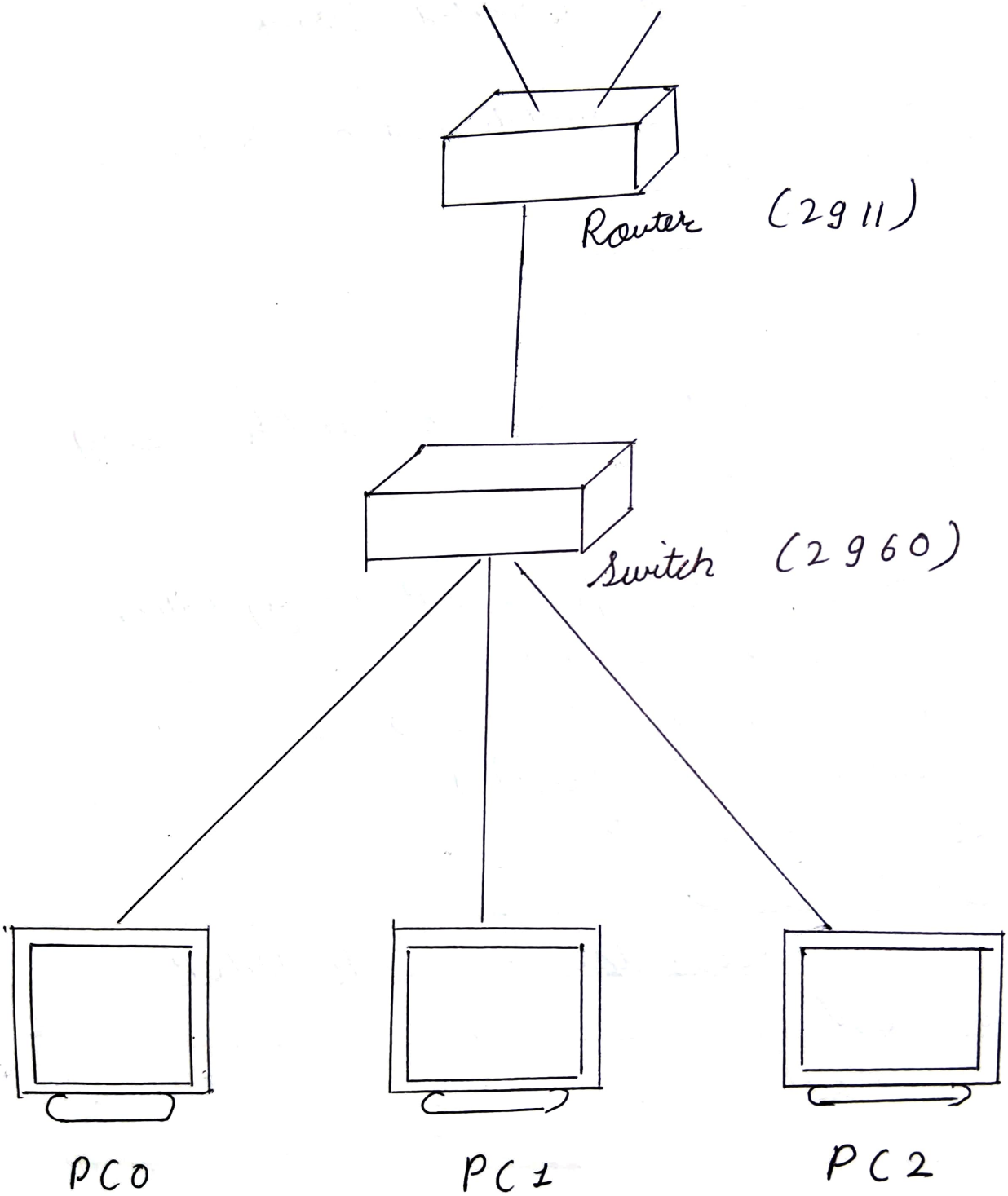

Fig :- DHCP using Router

Date

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- Maximum Users : 50

Click on Save

5 Configure PCs

Go to each PC: Desktop → IP Configuration → DHCP

Result :-

All PCs receive IP Addresses automatically from DHCP server

PART - B : DHCP using Router.

Requirement :- 1 Router (2911), 1 Switch (2960), 3 PCs

Procedure:

1) Connect Devices

- Connect PCs → Switch → Router

2) Configure Router Interface

Go to Router CLI:

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
```

3) Exclude IP Range

```
ip dhcp excluded-address 192.168.2.1 192.168.2.9
```

4) Create DHCP Pool

```
ip dhcp pool serverPool
```

Teacher's Signature

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network 192.168.2.0 255.255.255.0
default-router 192.168.2.1
dns-server 8.8.8.8
exit

5 Configure PCs
Go to PC → Desktop → IP configuration → DHCP

Result :
All PCs receive IP addresses automatically from the router